

Formulating Absolute Value Linear Equations

TEACHER KEY | Algebra II | Unit 2 | Day 4 | TEK A2.6.D

Objective: I can formulate (set up, without solving) an absolute value linear equation from a real-world description.

Vocabulary

Formulate — set up an equation from a description - **do not solve it yet.**

Center value — the target/ideal number the situation is measured against.

Margin — how far off from the center is still acceptable.

Template — every problem in this lesson fits the pattern $|x - \text{center}| = \text{margin}$.

Part 1 — Clarissa's New House

Clarissa wants a house within 5 houses of house #1248 on an even-numbered street.

Center value: **1248** Margin: **5**

Equation: **$|x - 1248| = 5$**

Part 2 — Surveying Margin of Error

A surveying instrument measures a 10-mile stretch with a 1.5% margin of error.

The 1.5% margin of error is measured as a percent of the true 10-mile distance x , so it is written $1.5 = |x - 10|/10 \cdot 100$.

Equation: **$1.5 = |x - 10|/10 \cdot 100$**

Part 3 — Brighton Builders' Hallway

A hallway has offices on both sides, evenly spaced from the center at $15x$ feet, and the total layout must equal 292 feet with 52 feet reserved for walls and doors.

Equation: **$2|15x| + 52 = 292$**

Practice — Formulate, Do Not Solve

1. A recipe calls for 2 cups of flour, with acceptable measurement error of 0.25 cups.

Center: **2** Margin: **0.25** Equation: **$|x - 2| = 0.25$**

2. A shipping company's box-weight target is 50 lb, with a tolerance of 3 lb.

Equation: **$|x - 50| = 3$**

3. A factory's bolt length must be within 0.02 cm of 6 cm.

Equation: **$|x - 6| = 0.02$**

Exit Ticket

A thermostat targets 70°F, with an acceptable range of 2°F above or below. Formulate an equation - do not solve.

Center: **70** Margin: **2**

Equation: **$|x - 70| = 2$**